AILEIKE Micro-Pressure Calibration Bench ALKT511

Technical White Paper

1. Product Overview and Technical Background

The AILEIKE Micro-Pressure Calibration Bench ALKT511 is a portable pressure generation and regulation device specifically designed for the calibration of micro-pressure instruments. It can output positive and negative pressure signals within the range of -100 to 600 kPa, making it suitable for laboratory verification and field calibration of micro-pressure instruments such as pressure gauges, pressure transmitters, pressure sensors, and pressure switches. The product features simple operation, smooth pressure ramping, excellent sealing performance, and ease of maintenance. It balances accuracy with portability to meet micro-pressure calibration needs across multiple scenarios.

2. Technical Specifications and Performance Metrics

2.1 Basic Technical Specifications

Parameter Category	Technical Specifications	Notes
Operating Environment	Laboratory or Field Use	Suitable for various working environments
Ambient Temperature	-10 to 60°C	Wide operating temperature range
Relative Humidity	<80%	Moisture-resistant design
Pressure Range	-100 to 600 kPa	Supports both positive and negative pressure requirements
Measured Medium	Clean air	Excellent compatibility with various media
Pressure Output	2	Can connect both standard meters and test meters simultaneously
Threaded Connection	M20×1.5 or as specified by the user	Flexible interface configuration
Adjustment Precision	min 10 Pa	High-precision pressure regulation
Dimensions	320×190×140 mm	Compact design
Weight	4.6 kg	Easy to carry

2.2 Key Performance Features

High-Precision Adjustment: With a sensitivity as low as 10 Pa, it can precisely reproduce low-pressure calibration points, ensuring the accuracy and reliability of calibration data;

Wide Measurement Range: The measurement range spans -100 to 600 kPa, integrating both positive and negative pressure generation into a single unit. The convenient switching between positive and negative pressure accommodates a variety of pressure testing and calibration needs.

User-Friendly Design: Equipped with dual standard output interfaces and an open-frame design, it enables quick connections and intuitive adjustments, significantly reducing the learning curve for operators;

Hygienic Operation: Air is used as the pressure transmission medium, ensuring a clean and hygienic environment without contamination, which also helps improve the accuracy of calibrated instruments.

Pressure Control: Utilizing advanced sealing technology, it offers high pressure control accuracy and smooth pressure ramp-up, providing a reliable pressure source for pressure testing and calibration tasks.

3. Main Application Areas

Based on its core advantages of precision and portability, the ALKT511 Micro-Pressure Calibration Bench can be widely applied across multiple fields:

Metrology and Testing Institutions: Used for laboratory precision verification of instruments such as pressure gauges and pressure transmitters, in compliance with metrological calibration standards;

Industrial Production Sites: Regular calibration and maintenance of pressure instruments on production lines in industries such as petrochemicals, power generation, and pharmaceuticals;

Laboratory Settings: Simulation and testing of low-pressure environments, providing a stable gas pressure source for research projects;

Instrument Manufacturing Industry: Serves as factory acceptance testing equipment for pressure instruments during production, ensuring product quality;

Research and Education: Used in laboratories at universities and research institutes for research and teaching experiments related to pressure measurement.

4. Operation and Maintenance Guide

Operating Procedure

Positive Pressure Testing

Unscrew the plug from the outlet connector and turn the fine-adjustment knob outward. Then, install the positive pressure standard gauge and the gauge under test on the two pressure output ports, respectively. Open the pressure-stabilizing valve and close the relief valve. Use the fine-adjustment knob to turn inward and increase the pressure to approximately the calibration point. Stabilize for 1–2 minutes; once the standard gauge reading stabilizes, test point by point up to full scale. After calibration is complete, relieve the pressure before removing the meter under test and the standard gauge.

If the reading slowly drops during pressurization (indicating a slight leak), close the pressure-stabilizing valve.

Negative Pressure Testing

Unscrew the plug from the outlet fitting and turn the fine-adjustment knob inward (stop when it cannot be turned further). Install the negative pressure standard gauge and the meter under test on the two pressure output ports, close the relief valve, and open the pressure-stabilizing valve. Slowly turn the fine-adjustment knob outward until it reaches approximately the calibration point, then hold it steady for 1–2 minutes. Once the reading on the standard gauge stabilizes, proceed to test each point up to full scale. After calibration is complete, open the pressure relief valve. Only after the pressure has been released may the test instrument and standard gauge be removed.

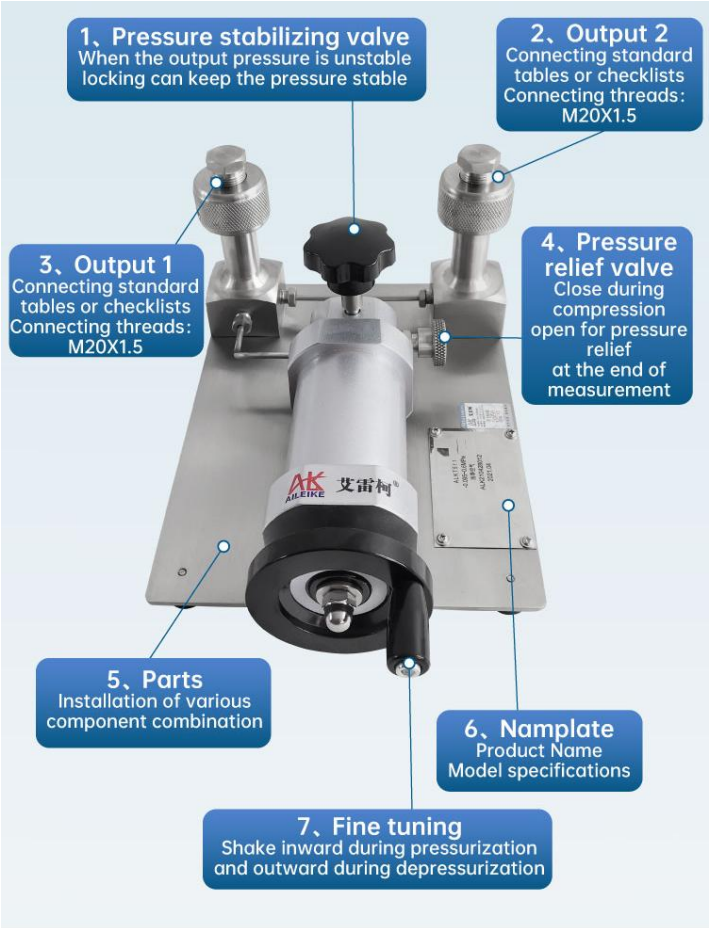
Maintenance Guide

Place the calibration bench on a flat, easily accessible surface and avoid using it in environments with highly corrosive gases or dust particles. The instrument under test should be properly cleaned to prevent contaminants from entering the test circuit.

The outlet connector is a dedicated quick-connect fitting. When installing the instrument, apply even pressure to tighten it without the need for tools. Once the sealing surfaces are in close contact, tighten it an additional 1–2 turns.

The outlet connector consists of two parts: an upper and a lower section. During use, frequently check whether the lower locking screw is securely fastened. If the connection is unreliable, it may come loose under high pressure.

5. Structure and Function



6.Conclusion

The ALKT511 Micro-Pressure Calibration Bench is a high-quality measurement and control device developed by AILEIKE based on industry needs and technical expertise. With 10 Pa resolution, wide measurement range coverage, user-friendly operation, and stable, durable performance as its core strengths, it effectively addresses the requirements for precision and convenience in the field of micro-pressure calibration. The product is widely applicable across various industry scenarios, providing users with efficient and reliable micro-pressure calibration solutions.